

Subconscious Learning's influence on employee development in knowledge-based markets

Virgilio M F Almeida

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Sadia Riaz

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Introduction

Background and problem

Formal training remains a cornerstone of workplace development, yet it often struggles to keep pace with the rapid changes in tools, systems, and demands characteristic of knowledge-based sectors (CIPD, 2020, 2023). Employees frequently report that much of their real capability develops through everyday tasks, informal conversations, and problem-solving rather than those structured courses that many companies pay for (Eraut, 2000). This style of informal learning is often described as tacit or non-formal, which mostly happens through action and experience rather than deliberate study and concentrated memorisation (Eraut, 2000).

The challenge lies in understanding *how* this informal capability develops. Psychological research suggests that humans possess subconscious learning mechanisms – such as implicit pattern recognition and statistical learning. That those processes operate below full awareness and without deliberate intention (Reber, 1996; Saffran et al., 1996). However, these mechanisms have not been clearly linked to workplace contexts nor integrated with practical theories of professional development.

Definition

As explained by Kaufman et al. in 2010, implicit learning can be defined as “Learning without deliberate intention or full awareness of what has been learned” making it a direct subcategory of subconscious learning. He argues it is partly independent from IQ and working memory but linked to performance in complex tasks (Kaufman et al., 2010).

American Psychological Association (2018) noted that “Implicit learning is evidenced by improved task performance rather than as a response to an explicit request to remember.”

Subconscious learning, in relation to the workplace, can be seen when employees take in patterns, information, or ways of working from their environment and later use them, without deliberately trying to learn and often without realising exactly what they have learned. This can be seen through tacit learning and human interactions while in the workplace.

Research gap

While the literature on workplace learning highlights the importance of learning cultures and informal processes (CIPD, 2020), it lacks a specific framework for the subconscious processes that may underpin tacit capability growth. Tacit knowledge is the know-how that guides action that can't be fully expressed (Eraut, 2000). Similarly, cognitive theories of implicit learning have yet to be applied systematically to real work environments. (Gourlay, 2002)

This gap matters because organisations need actionable frameworks to support capability development beyond formal training, particularly in dynamic knowledge work where formal instruction cannot cover every scenario or change.

Aims and objectives

This study aims to explore how subconscious learning contributes to capability development in knowledge-based work. By review cognitive and workplace learning literature relevant to subconscious processes a working theory for the subconscious adaptability will be developed in workplace contexts. This understanding will be tested and applied to other parts of the world, beginning with the EU as an example.

The main question is, to what extent do employees in knowledge-based work develop their capabilities through subconscious learning.

Literature Review

Subconscious learning in workplace

Subconscious learning is commonly framed through implicit learning, statistical learning, and tacit knowledge, yet the literature does not treat these as identical processes. Reber (1996) defines implicit learning as the acquisition of complex structure that occurs “largely independently of conscious attempts to learn” and outside explicit awareness. That claim becomes concrete in Saffran et al. (1996), where 8-month-old infants segmented speech after only 2 minutes of exposure by tracking statistical regularities in continuous input. The significance of that finding lies not simply in its developmental novelty, but in what it suggests about the mind’s appetite for structure: repeated exposure can be converted into usable knowledge before the learner can explain how the conversion happened. Alamia et al. (2016) complicate this further by showing that unconscious associative learning can occur even when cues are consciously visible, which weakens any neat assumption that awareness of input guarantees awareness of learning. Subconscious learning has become less of a single mechanism and more of multiple processes acting together.

Kaufman et al. (2010) found implicit learning to be “unrelated to working memory” and only weakly tied to psychometric intelligence, yet it is still associated with verbal analogical reasoning, processing speed, and foreign-language exam performance. It resists the easy temptation to collapse subconscious learning into standard IQ or memory explanations, while

also refusing to cast it as a marginal skill with little practical value. Montuori and Montefiori (2022) extend this line by treating implicit learning as a possible cognitive ability in workplace contexts, where employability and performance depend on more than what workers can consciously report or deliberately rehearse. One might say that the later paper moves the first one from the laboratory to the office floor. Still, the relationship between the two is not entirely smooth. The 2010 study leaves implicit learning partly distinct, but not fully separate, from broader reasoning ability; the 2022 review leans harder toward application yet does not erase that ambiguity. That friction is useful, because it prevents subconscious learning from being treated as either a mystical force or a fully independent talent.

The tacit knowledge literature gives this same issue a more practical vocabulary. Polanyi's famous idea that "we can know more than we can tell," as summarised by Gourlay (2002), captures the basic tension: some knowledge guides action without ever becoming fully verbalised. Eraut (2000) develops that point by arguing that professional learning is not just deliberate and planned, but also reactive and implicit. In his account, much of the learning that matters in work happens during ordinary activity, through judgement, coping, and repeated exposure to changing situations. Smith (2001) takes a related but slightly different line. Where Eraut emphasises the process of learning through work, Smith draws a sharper distinction between explicit knowledge, which can be documented and taught, and tacit know-how, which sits inside action, perception, and experience. The two accounts are aligned, but they do not say exactly the same thing. Eraut shows how tacit learning develops in practice; Smith shows why organisations struggle to capture it in formal systems. Together, they suggest that professional competence is built on more than procedures.

That underlayer becomes even clearer in the work on unconscious integration and statistical learning. Kuldas et al. (2013) argue that conscious integration of verbal and pictorial

materials does not always occur, because the learner's mind cannot keep every piece of information in deliberate focus at once. Their point is not that people fail to understand, but that understanding often emerges from processes that never become fully conscious. Smalle et al. (2022) add an important wrinkle. They show that adults still possess strong implicit statistical learning abilities, yet these abilities can be masked or altered by cognitive control. This introduces a productive tension: on one hand, subconscious learning appears robust across the lifespan; on the other, it may be suppressed by overcontrol, over monitoring, or the wrong instructional conditions. The implication is subtle. Subconscious learning does not disappear in adulthood, but it may work best when cognition is not being forced into constant self-inspection.

Organisational learning research, which usually concentrates on visible routines, formal training, and deliberate adaptation. Yet the field often privileges the parts of learning that are easiest to document, which leaves implicit and tacit processes in the shadows. CIPD's Learning at Work 2023 survey is useful here because it shows that the workplace has not settled on a single training model. The report finds that digital learning continues to rise, with 48% of learning practitioners reporting an increase in use, while only 25% of organisations now say learning is delivered entirely face-to-face and an identical 25% say it is delivered entirely online or digitally (CIPD, 2023). At the same time, there has been a renewed use of more traditional interventions, including formal qualifications, apprenticeships, job rotation, secondment, and shadowing (CIPD, 2023). That pattern doesn't show that formal and informal methods are in opposing camps. Instead, modern training is already mixed, which leaves room for knowledge that can develop below the level of explicit instruction. The formal system may provide the frame, but tacit and implicit learning often do the heavy lifting inside it.

European Training contexts

European countries differ systematically in training approaches, affecting how much learning occurs through work immersion versus formal instruction (Cedefop, 2020, 2023).

Countries can be divided into 5 areas of the EU based on their location and culture.

The North

The countries that can categorise “The North” would be places such as Germany, Austria or Switzerland. These countries are categorised by their use of dual systems with strong work-based apprenticeships; learning embedded in real tasks. (Cedefop, 2024) As Bliem et al. (2017) argue, this region is grouped not merely by geography but by a shared "training logic" where workplace learning is the primary pedagogical driver. In this model, apprenticeships are not a supplement to education but the core of it. Evidence from Cedefop (2024) shows that in Austria, apprentices spend approximately 80% of their time in companies, a figure mirrored in Switzerland where the Swiss Confederation (2025) highlights a three-to-four-day weekly immersion in corporate environments. This could mean that vocational identity is forged through lots of structured participation in the workforce, with more formal schooling being a theoretical complement to the practical experience.

The East

Eastern Europe, particularly Poland and Romania, represents a transitional model where training is increasingly driven by market pressure and the demands of the global digital economy. As TTaaS (2021) and Zartis (2023) argue, these nations have emerged as "IT

powerhouses" where learning is characterized by "cutting-edge projects" and active problem-solving rather than traditional apprenticeships. In this context, competence is gained through the intensity of the work task itself—often within the outsourcing sector—where rapid skill acquisition is a necessity. This project-driven approach reflects an economic structure that values technical delivery and "21st-century skills" over long-term institutionalized training pathways (EduBot, 2023).

The South

Southern Europe is composed of Mediterranean countries such as Italy, Spain, and Portugal. The training architecture around these countries combines formal structures with a heavy reliance on informal learning within Small and Medium Enterprises (SMEs). In Italy, for instance, Cedefop (2025) notes that a staggering 86.1% of adults participate in informal learning, indicating that the workplace is the dominant site for skill acquisition, even when formal participation is lower. Spain and Portugal have responded to this reality by building "experience-based competence" into their legal frameworks. Thus, the Southern model suggests a pragmatic synthesis where the state officially recognizes the organic learning that occurs within less structured SME environments.

The West

Western Europe could be exemplified by places like the UK or France, where it favours a "platform-centred" approach that emphasizes modular, digital, and employer-led development. The argument for this region centres on speed and scalability; corporate platforms like Deloitte's FinLearn (2024) use CPD-accredited e-learning to help professionals

stay current in fast-evolving sectors like finance. As Learning Light (2025) suggests, the goal is often to reduce "time to competency" through simulation and accelerated induction. While the D-A-CH region produces competence through direct workplace participation, the Western model leverages technology to package work-oriented learning into modular, self-directed formats.

The Nordics

Nordic countries like Denmark, Sweden and Finland, prioritize a model of "lifelong learning" that emphasizes the portability of skills across the life course. Unlike the Northern apprenticeship model, the Nordic approach focuses on making prior experience "legible" through formal validation. According to the Nordic Network for Lifelong Learning (2025), training is viewed as a continuous process that moves fluidly between school, work, and digital environments. This is supported by Cedefop's (2024) definition of validation as a systematic process of "identification, documentation, [and] assessment." The Nordic region fosters a culture of flexibility and reskilling that distinguishes it from more rigid initial training routes.

Subconscious Adaptive Learning Theory

This study proposes the Subconscious Adaptive Learning Theory (aka. SALT) as a theory attempting to mix cognitive and workplace perspectives. SALT is the environment-driven process by which workers, through repeated participation in focused tasks and social tasks, extract patterns, where they assign emotions to assemble schemas for competent performance.

SALT foregrounds subconscious mechanisms as the engine of tacit workplace learning, scoped to dynamic knowledge work where formal training lags. It complements Eraut's (2000) non-formal learning and CIPD's (2020) learning cultures by specifying *how* exposure changes to capability. While SALT wouldn't replace formal training, it should explain gap-filling in complex, changing environments, allowing it to be used along any culture.

SALT across European contexts

Region	SALT implications
North	Dual immersion maximises pattern exposure for SALT.
West	Mixed systems; SALT fills e-learning gaps.
Nordic	Flexibility supports SALT through autonomy.
South	SMEs rely on SALT despite school focus.
East	Project work creates SALT opportunities in IT.

(Table 1)

Methodology

This project uses a mixed method approach combining a longitudinal participant experiment along with a coded simulation. 2 participants (P1 & P2) were asked to complete a five-week online experiment as junior analysts in a simulated UK financial services firm, mirroring real-world tasks to probe subconscious learning through repeated exposure to escalating cognitive demands. A parallel agent-based simulation with 50 agents reproduced these tasks exactly, operationalising latent traits like rule-following and pattern-detection to model path-dependent adaptation under controlled noise.

Experiment

Participants were assigned the role of junior analysts in a UK-based financial and technological environment. This role was selected because it required the coordinated use of quantitative reasoning, rule application, table interpretation, and many other skills. The junior analyst persona provided a controlled yet ecologically valid framework for assessing multi-modal task performance. The experiment was delivered online across 15 sessions, organised as three sessions per week, with baseline testing conducted before the main sequence and additional measurement points at Weeks 2, 4, and 5.

Week 1 served as the baseline phase and was characterised by binary classification, threshold application, and low-ambiguity retrieval. Week 2 introduced additional conditional logic, aggregation, and multi-table comparison. Week 3 increased formal reasoning demands through equation use, proportional scoring, and transfer to modified rules. Week 4 required integration across multiple financial indices and contextual cues, while Week 5 imposed the highest load through full advisory judgement, multi-criteria interpretation, and rule critique.

This escalation was intentional and was treated analytically as a potential confound because changes in performance could reflect both learning and task complexity escalation.

Task structure:

Each week contained three tasks. Task 1 focused on structured rule following and quantitative classification. Task 2, on the other hand, emphasised business interpretation, trend extraction, and recommendation formation from tabular or graph-like data. Meanwhile, Task 3 targeted anomaly detection, conceptual debugging, and rule auditing. The recurring three-task format supported longitudinal comparability while preserving variation in surface content.

On the 1st task of Week 1, participants applied a binary client-flagging rule to a small table and answered follow-up questions on borderline cases, revised rules, and confidence calibration. This task indexed threshold sensitivity, rule adherence, and metacognitive appraisal. In Task 2, participants were focused on extracting values from traffic-source and page-performance tables and summarised change in overall website performance. This indexed metric extraction, comparison accuracy, and verbal justification. In Task 3, participants identified anomalous rows and distinguished increasing procedural efficiency from continued stepwise processing. This indexed pattern detection and early signs of automatised judgement.

Week 2 increased conditional complexity and arithmetic demand. Task 1 required application of multiple flagging criteria across account records and estimation of averages and total fees. Task 2 combined account classification with user-growth metrics, churn estimates, conversion rates, and feature-rollout recommendation. Task 3 required prioritisation of

incidents using severity, resolution time, recurrence, and system area, alongside commentary on temporal pressure trends.

Week 3 constituted the first substantial transition to formalised reasoning. Task 1 introduced proportional risk scoring, ranking, and transfer to revised rule structures. Task 2 required forecast generation from trend tables, cost information, and contradictory indicators. Task 3 required parsing of conditional logic, distinction between conjunction and disjunction, and identification of rule malfunction. This phase assessed the extent to which repeated exposure had supported transfer beyond rote application.

Week 4 represented a high-load integrative phase. Task 1 required the joint use of PRS, VAR, BRS, SS, BS, and FRI to rank clients, interpret structural versus behavioural risk, and identify contradictory financial profiles. Task 2 required interpretation of stock-price trajectories, volume dynamics, sector context, and event information to generate forecasts and portfolio recommendations. Task 3 asked the participants to compare system outputs with rule logic and fundamentals, followed by identification of rule deficiencies and candidate revisions.

Week 5 constituted the maximal-load phase. Task 1 presented full client case files including statement data, debt structure, arrears, behavioural indicators, and banking actions, and required a funding recommendation. Task 2 required portfolio allocation across the S&P 500, gold, and “Stock X” using historical price paths, volatility descriptors and yield characteristics within a macro context of the world. Lastly, Task 3 required application of multiple rules to the same asset table, identification of rule flaws, proposal of code revision, and production of final human recommendations.

(Stock X is based on a real stock)

Measurement and outputs:

Weekly performance was summarised using mean trajectories, standard deviations, minimum and maximum values, and percentage change across time. Mean-line graphs were used to describe longitudinal change, while descriptive tables captured dispersion and the direction of movement.

Simulation

The simulation operationalised the same weekly structure with 50 agents. Each agent was assigned a trait profile, dynamic state variables, and error parameters. The simulation preserved task isomorphism with the human study: identical prompt logic, equivalent scoring rules, and comparable weekly escalation were implemented.

Disclaimer:

The simulation includes randomness because not every agent responds identically, even when given the same task structure. This randomness represents human variability, attention shifts, and inconsistent decision making. The model assumes that weekly trait changes can be approximated using repeated update rules, that the task structure remains stable across weeks, and that the selected traits are sufficient to represent the main learning process. It also assumes that the coded agents are a useful proxy for participant behaviour, even though they are simplified versions of real people.

The model assumes that the experiment measures subconscious learning through repeated exposure and gradual change. It also assumes that the weekly tables and graphs can be interpreted as trends in learning rather than isolated events.

Initial agent stats:

The agents were given their initial scores randomly, using the participant's scores from the self-assessment ± 2 so that we can assume all agents (which would represent humans) have roughly the same skill level as the participants

Parameters:

The simulation starts from the same initial values as the experiment dataset. Each agent is assigned values for the same core traits and error measures. Weekly updates are then applied according to the model rules. Some traits are allowed to improve more strongly in the early weeks, while later weeks may show slower gains, plateaus, or decline depending on the trait and the error structure.

The simulation used three parameter layers: stable participant-like traits, dynamic weekly states, and error parameters.

Layer 1: Stable participant traits

General cognitive traits

Parameter	Meaning
rule_following	How consistently the participant applies explicit rules
arithmetic_fluency	Speed and reliability with basic calculations
equation_fluency	Ability to use formulas correctly
table_integration	Ability to combine information across rows/tables
pattern_detection	Ability to spot anomalies, trends, and unusual cases
ranking_skill	Ability to compare items and order them correctly
verbal_justification	Ability to explain choices clearly in short written form
working_memory	Ability to hold multiple constraints in mind at once
noise_tolerance	Ability to stay accurate under clutter and many fields
ambiguity_tolerance	Ability to function when the “right” answer is not perfectly explicit

(Table 2)

Note: The simulation used three parameter layers: stable participant-like traits, dynamic weekly states, and error parameters.

Domain-style traits

Parameter	Meaning
business_intuition	Ability to infer what matters commercially, beyond the literal numbers
forecasting_style	Tendency toward conservative vs optimistic forecasting
trend_sensitivity	How strongly the participant reacts to recent trends

contradiction_detection	Ability to notice when different indicators point in different directions
risk_appetite	Preference for safer vs more aggressive recommendations
automation_trust	Tendency to trust a system label/output by default
threshold_sensitivity	Ability to notice and respect exact threshold boundaries
override_willingness	Willingness to disagree with a rule or system output when it seems wrong
conceptual_debugging	Ability to explain why a rule fails at the conceptual level, not just the row level

(Table 3)

Note: These were included because the later experiment weeks require more than raw cognition; they also require applied judgement, willingness to challenge system output, and sensitivity to contextual or contradictory evidence. Humans can naturally process these things without realising when judging a task and finding an outcome.

Learning-style traits

Parameter	Meaning
learning_rate	How quickly the participant improves with repeated exposure
chunking_ability	Ability to move from step-by-step processing to pattern-based processing
transfer_ability	Ability to apply earlier learning to a changed task
confidence_calibration	Whether confidence tracks real accuracy or not

(Table 4)

Note: These were especially important for a subconscious learning study because they operationalised how quickly an agent improves, whether it shifts from step-by-step to pattern-based processing, whether it can transfer earlier learning to a changed task, and whether confidence tracks actual ability.

Layer 2: Dynamic weekly states

Parameter	Meaning
fatigue_t	Temporary performance drag at week t
confidence_t	Current subjective confidence at week t
familiarity_task1_t	Growing familiarity with Task 1-style structure
familiarity_task2_t	Growing familiarity with Task 2-style structure
familiarity_task3_t	Growing familiarity with Task 3-style structure
overload_t	How close the participant is to cognitive overload that week
adaptation_t	How much the participant has learned to simplify or chunk the task

(Table 5)

Note: These states were designed so that the same agent could perform differently from week to week even without changing its stable core profile. This is methodologically important because real participants do not behave like fixed machines; performance depends partly on temporary cognitive state as well as trait-level ability. Some people may do great in one week but not as much on the next.

Layer 3: Error parameters

Parameter	Meaning	Example consequence

slip_rate	Chance of making a small accidental error despite knowing the rule	Miscounts one flagged row
threshold_miss_rate	Chance of mishandling a boundary condition	Treats $\geq$ as $>$
omission_rate	Chance of forgetting one condition in a multi-condition rule	Applies only half of the rule
arithmetic_drift	Tendency for rough calculations to drift off	Slightly wrong estimated totals or averages
formula_mixup_rate	Chance of confusing formulas or variables	Uses PRS logic when VAR was needed
ranking_noise	Chance of misordering near-tied cases	Gets top two reversed
explanation_weakness	Tendency to reach a decent answer but justify it poorly	Right client, weak reason
contradiction_blindness	Chance of missing signals that pull in opposite directions	Misses strong margin but bad leverage conflict
automation_anchor	Tendency to stick too closely to the system's output	Fails to override wrong tag
recency_bias	Tendency to overweight the most recent observation	Overreacts to the latest week or price move
salience_bias	Tendency to overweight the most visually obvious cue	Focuses on one red flag and misses the structure

(Table 6)

Note: These gave the model psychologically plausible failure modes. Instead of representing low performance as a single undifferentiated weakness, the simulation represented different kinds of

mistakes that map onto different human-like failures, such as forgetting one part of a multi-condition rule, overreacting to the latest trend, or trusting a system output too easily, allowing for the variable of “human error” in the code.

Agent responses were scored using task-specific sub-scores rather than a single global mark. Weekly task scores were then used to update agent traits through incremental learning rules. High performance produced modest upward adjustment in relevant traits, whereas weak performance, overload, or error accumulation attenuated later gains. The update process thus represented a path-dependent learning dynamic.

Sub-score	Meaning
flagging_accuracy	Correct application of binary or categorical rules
arithmetic_accuracy	Accuracy of calculations and estimates
equation_accuracy	Correct use of formulas
ranking_accuracy	Correct ordering/comparison
anomaly_accuracy	Spotting suspicious or unusual records
forecast_quality	Producing sensible future estimates
interpretation_quality	Correct reading of what numbers imply
recommendation_quality	Choosing and defending a plausible action
audit_accuracy	Correctly spotting wrong system outputs
rule_revision_quality	Quality of proposed rule/code improvements

(Table 7)

Data Analysis

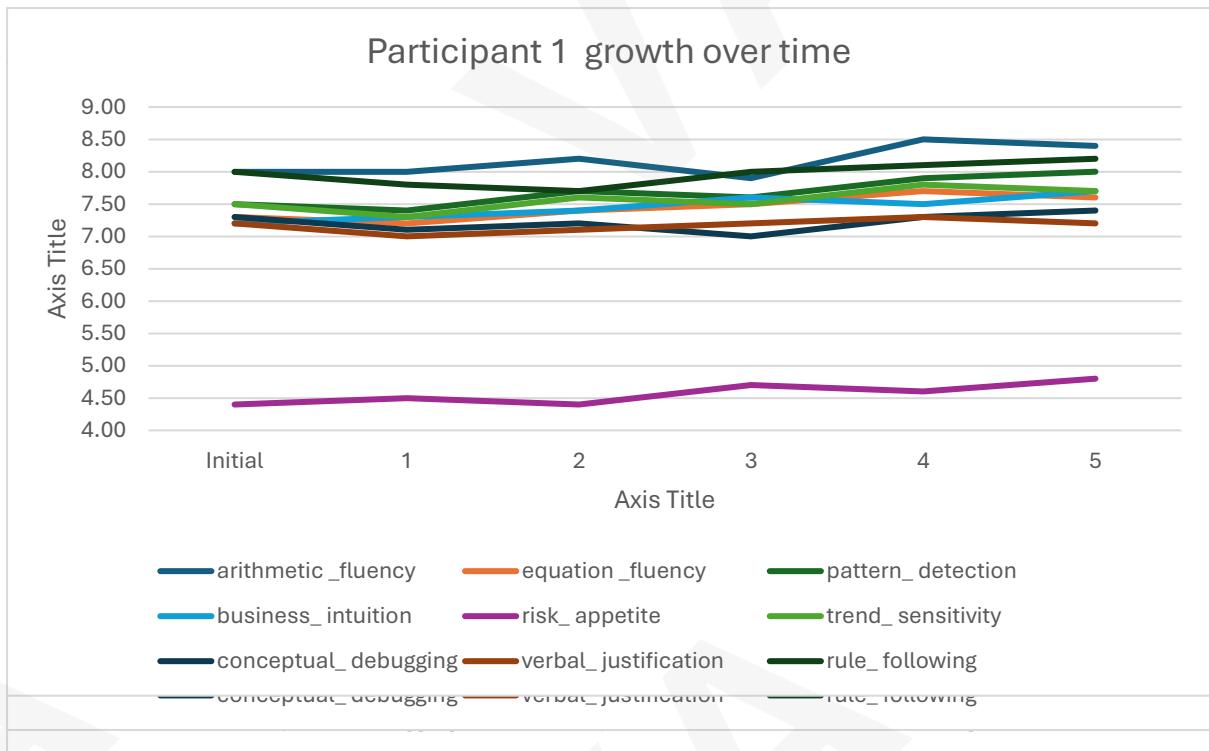
Participant Results

Parameter	P1 initial	P2 initial	Difference (P1 - P2)
Calculation skill	8.0	6.3	1.7
Forecasting / pattern skill	7.5	5.5	2.0
Code / logic understanding	7.3	5.8	1.5
Noise tolerance (overloaded info)	6.8	4.8	2.0
Motivation / engagement	7.2	5.8	1.4
Fatigue / overload sensitivity	4.4	4.8	-0.4

(Table 8.5)

Note: These results come from the self-assessment, therefore believed to have personal bias. Were used as the anchors for designing the initial “stats” of the agents.

(Figure 1.1)



(Figure 1.2)

Agent results

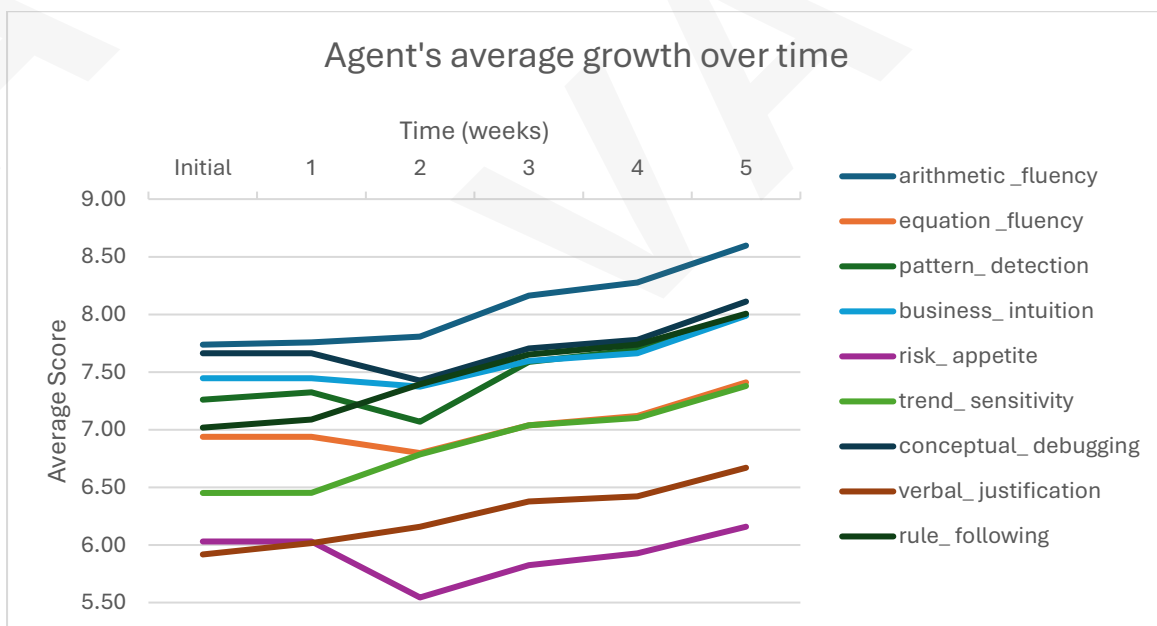
The agent results show steady improvement across the repeated measures. For arithmetic fluency, the mean rose from 7.7372 at Initial to 8.5967 at Week 5 (Figure 3.1), while the paired t-test moved from non-significant change in Week 1 [$t = -0.91, p = 0.36$] (Figure 2.1) to a highly significant increase by Week 5 [$t = -15.06, p = 5.13 \times 10^{-20}$] (Figure 3.1). The same pattern appears in equation fluency, business intuition, and trend sensitivity, where Week 1 changes were small (Figure 2), but Week 5 gains were clear and statistically strong.

Measure	Initial Mean	Week 5 Mean	Change	Week 5 p-value
Arithmetic fluency	7.7372	8.5967	+0.8595	5.13×10^{-20}

Measure	Initial Mean	Week 5 Mean	Change	Week 5 p-value
Equation fluency	6.9370	7.4100	+0.4730	5.99×10^{-12}
Business intuition	7.4456	7.9883	+0.5427	1.97×10^{-9}
Trend sensitivity	6.4508	7.3791	+0.9283	4.12×10^{-22}

(Table 11)

Note: The t-statistics were strongly negative because the Week 5 means were higher than the initial means, and the p-values were extremely small, which shows that the improvement was statistically significant



(Figure 2)

Note: The growth rate is not constant across the full period. It is stronger in the early and middle weeks, then slows down slightly in Weeks 4 and 5, which is exactly what you would expect when the agents are consolidating earlier gains rather than learning at the same speed throughout.

Discussion

SALT's empirical test shows results that can be seen as both encouraging and disheartening. The agent simulation demonstrates that through repeated exposure to tasks, a steady and measurable increase can be noticed in skills that define knowledge work. Things like spotting patterns in noisy data or finding and applying rules under pressure, and even forming sensible judgments without direct guidance from someone. In the simulation, certain traits like arithmetic fluency and trend sensitivity showed consistent improvement across the five weeks (effect size = 2.12 & 2.39 respectively), with the participants effectively calculating simple binary decisions from Week 1 into the more integrative advisory tasks of Week 5 (Figure 4). Supporting the core claim, that subconscious processes can turn environmental signals into usable scripts for performance, just like Saffran et al. (1996) showed infants extracting word boundaries from statistical regularities in speech streams. The agents didn't need lots of time to think. Their updates happened through rules in the code that mirrored how repeated practice might build other skills in a subconscious manner. It's a clear demonstration of the theory's potential, especially in a context like junior financial analysis, where tasks like client flagging or portfolio analysis rely on mixing numbers, rules, and intuition in ways that formal training rarely covers.

That said, the human participants tell a more complicated story. P1 and P2 made progress, with P2's forecasting skill, for instance, improving noticeably from baseline (trend_sensitivity = 12.73% growth). However, their trajectories were far less smooth than the agents', with plateaus in noise tolerance and verbal justification that suggest fatigue and overload played a bigger role than the simulation could capture (Figure 1.1, Figure 1.2). Week 5's multi-metric client prioritization in task 1 and portfolio decisions in task 2 demanded not just pattern recognition but also the emotional tagging that comes with it, yet the participants' responses

stayed cautious, favouring a safer S&P/Gold allocation over riskier but more profitable Stock X. This "human-agent chasm," is the study's most revealing finding. It suggests SALT works best when the learner is unburdened by real-world drags. As participants also had their lives to care for or the pressure to explain choices in the experiment. This aligns with Kaufman et al.'s (2010) view of the cognitive unconscious as a structure-inducting system, operating independently of working memory or IQ-linked effort. This does raise a question: if subconscious learning is so potent, why didn't the humans close the gap? The answer may lie in the simulation's abstraction as agents lack the fatigue states or recency biases in their modelled errors.

This performance divergence forces the confrontation in the literature that SALT was meant to bridge. On one hand, the cognitive side, with researchers like Saffran et al. (1996), Kaufman et al. (2010), Kuldass et al. (2013), paints subconscious learning as robust and automatic, capable of handling complex structures without deliberate intent. Kaufman's work, disconnects it from psychometric g (weak $r < 0.2$), tying it instead to real-world skills like analogical reasoning and processing speed. On the other hand, the tacit knowledge tradition, with researchers like Polanyi, Gourlay (2003), Eraut (2000), complicates the picture by emphasizing that much workplace learning is reactive and embodied, emerging from coping in messy, social contexts rather than pure pattern recognition and application. Eraut (2000) argues that implicit processes thrive in "ordinary activity" like judgment under time pressure, but they demand emotional tagging and collective feedback that the simulation omitted. Gourlay (2003) goes further, questioning whether "tacit knowing" is even a coherent category or just a label for what we can't yet articulate, he states "it is a concept without consistency, or clear foundation" (Gourlay, 2003). A perspective that the human justifications ("cautious long-term growth via diversified ETF") seem to echo, as they grasped the patterns but struggled to say them fluidly. SALT tries to split the difference, positing environment-driven

extraction as the engine, but the results suggest it's more theoretical than proven. The agents support the mechanism's functionality, while humans show how it's still vulnerable to cognitive friction and human error.

Smalle et al. (2022) offer a different perspective that cuts against the optimism of the agent data. They find adult statistical learning persists but gets "altered" by cognitive control, which is what may have happened when Week 3's PRS scoring shifted, where participants leaned on pre-learned thresholds rather than adapting. This "altering" effect disagrees with SALT's narrative, implying that over-monitoring (common in knowledge work audits) could suppress the chunking_ability the simulation showed. A view echoed in CIPD's 2023 survey, where 48% report digital learning growth but barriers like time and engagement persist, leaving informal tacit as the "heavy lifting" but hard to scale. The Warwick paper on tacit knowing (Gourlay, 2003) pushes back hardest, arguing tacit is often "behaving" under rules rather than deep unconscious structure.

When looking at Europe, the D-A-CH model's "dual vocational education and training systems" allows for apprentices to average "three to four days a week" training in firms (DC-dVET 2016) That means they conduct up to "80% of their training time in a company" in Austria (Cedefop 2024). Low youth unemployment, pegged below 10% across these nations versus EU-28's 22.2% (DC-dVET 2016). This pivots scrutiny to Nordics, where NOVA-Nordic (2023) charts "non-formal qualifications" levelled to NQFs via validation, legitimizing subconscious learning outputs in Denmark and Sweden's open frameworks, yet Norway's closure to non-formals exposes lifelong fluidity's uneven scaffolding.

East's outsourcing hubs note "high volumes of STEM graduates" and "cutting-edge projects" (TTaaS 2021), echoing Week 3 forecasts through project tacit buildup, but due to the relentless client intensity overload it can cause cognitive disruption rather than proper

learning if structured safeguarding is not placed. Damesin et al. spotlight "weaker" support for "on-the-job learning" reliance in southern regions like Italy and Spain (UIL/CEDEFOP). This lack of support causes trainees to not be fully immersed on the normal practices of the job, which are normally picked up without conscious awareness. Meanwhile, the west's e-platforms, via "system simulation" and "accelerated inductions" (Learning Light 2025), enable a steadier escalation to the job, short CPD modules (Deloitte 2024) limit deep subconscious learning due to not being able to pick up all factors of the full workplace.

Conclusion

Overall, subconscious learning can be seen as a plausible form of training for certain firms. As SALT provides a foundation for adult's learning capabilities, but the evidence here suggests that it is only part of the story. The agent simulation showed clear improvement through repeated exposure, especially in pattern detection, rule use, and moving from simple decisions toward more integrated judgement. That supports the idea that knowledge can be built through contact with structure, even without full conscious effort.

The human participants were less tidy. Whilst their improvement was visible, their progress was uncertain, shaped by fatigue, and pressure. Their Week 5 responses show that real workplace judgement is not just pattern recognition; it is also emotional restraint and risk management in uncertainty. The gap between the agents and the humans is therefore not a failure of the theory, but a reminder that subconscious learning operates inside messy contexts.

The wider literature fits this reading. With people like Kuldass, Gourlay and Eraut pointing towards knowledge that is partly hidden from awareness, yet the human data show that such

learning is not automatic in a clean sense. It appears to be strongest when the environment is controlled and repeated, and weakest when outside factors intervene.

Appendices

Tables:

Salt Regional table:

Region	SALT implications
North	Dual immersion maximises pattern exposure for SALT.
West	Mixed systems; SALT fills e-learning gaps.
Nordic	Flexibility supports SALT through autonomy.
South	SMEs rely on SALT despite school focus.
East	Project work creates SALT opportunities in IT.

(Table 1)

General cognitive traits Table

Parameter	Meaning
rule_following	How consistently the participant applies explicit rules
arithmetic_fluency	Speed and reliability with basic calculations
equation_fluency	Ability to use formulas correctly
table_integration	Ability to combine information across rows/tables
pattern_detection	Ability to spot anomalies, trends, and unusual cases
ranking_skill	Ability to compare items and order them correctly
verbal_justification	Ability to explain choices clearly in short written form
working_memory	Ability to hold multiple constraints in mind at once
noise_tolerance	Ability to stay accurate under clutter and many fields

ambiguity_tolerance	Ability to function when the “right” answer is not perfectly explicit
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(Table 2)

Domain-style traits Table

Parameter	Meaning
business_intuition	Ability to infer what matters commercially, beyond the literal numbers
forecasting_style	Tendency toward conservative vs optimistic forecasting
trend_sensitivity	How strongly the participant reacts to recent trends
contradiction_detection	Ability to notice when different indicators point in different directions
risk_appetite	Preference for safer vs more aggressive recommendations
automation_trust	Tendency to trust a system label/output by default
threshold_sensitivity	Ability to notice and respect exact threshold boundaries
override_willingness	Willingness to disagree with a rule or system output when it seems wrong
conceptual_debugging	Ability to explain why a rule fails at the conceptual level, not just the row level

(Table 3)

Learning-style traits Table

Parameter	Meaning
learning_rate	How quickly the participant improves with repeated exposure
chunking_ability	Ability to move from step-by-step processing to pattern-based processing
transfer_ability	Ability to apply earlier learning to a changed task
confidence_calibration	Whether confidence tracks real accuracy or not

(Table 4)

Dynamic weekly states table

Parameter	Meaning
fatigue_t	Temporary performance drag at week t
confidence_t	Current subjective confidence at week t
familiarity_task1_t	Growing familiarity with Task 1-style structure
familiarity_task2_t	Growing familiarity with Task 2-style structure
familiarity_task3_t	Growing familiarity with Task 3-style structure
overload_t	How close the participant is to cognitive overload that week
adaptation_t	How much the participant has learned to simplify or chunk the task

(Table 5)

Error parameters Table

Parameter	Meaning	Example consequence
slip_rate	Chance of making a small accidental error despite knowing the rule	Miscounts one flagged row
threshold_miss_rate	Chance of mishandling a boundary condition	Treats $\geq$ as $>$
omission_rate	Chance of forgetting one condition in a multi-condition rule	Applies only half of the rule
arithmetic_drift	Tendency for rough calculations to drift off	Slightly wrong estimated totals or averages
formula_mixup_rate	Chance of confusing formulas or variables	Uses PRS logic when VAR was needed
ranking_noise	Chance of misordering near-tied cases	Gets top two reversed
explanation_weakness	Tendency to reach a decent answer but justify it poorly	Right client, weak reason
contradiction_blindness	Chance of missing signals that pull in opposite directions	Misses strong margin but bad leverage conflict
automation_anchor	Tendency to stick too closely to the system's output	Fails to override wrong tag

recency_bias	Tendency to overweight the most recent observation	Overreacts to the latest week or price move
salience_bias	Tendency to overweight the most visually obvious cue	Focuses on one red flag and misses the structure

(Table 6)

Sub-score for agents table

Sub-score	Meaning
flagging_accuracy	Correct application of binary or categorical rules
arithmetic_accuracy	Accuracy of calculations and estimates
equation_accuracy	Correct use of formulas
ranking_accuracy	Correct ordering/comparison
anomaly_accuracy	Spotting suspicious or unusual records
forecast_quality	Producing sensible future estimates
interpretation_quality	Correct reading of what numbers imply
recommendation_quality	Choosing and defending a plausible action
audit_accuracy	Correctly spotting wrong system outputs
rule_revision_quality	Quality of proposed rule/code improvements

(Table 7)

Participant results of key stats

Agent	Week	arithmetic_fluency	equation_fluency	pattern_detection	business_intuition	risk_appetite	trend_sensitivity	conceptual_debugging	verbal_justification	rule_following
P1	Initial	8.00	7.30	7.50	7.20	4.40	7.50	7.30	7.20	8.00
	1	8.00	7.20	7.40	7.30	4.50	7.30	7.10	7.00	7.80
	2	8.20	7.40	7.70	7.40	4.40	7.60	7.20	7.10	7.70
	3	7.90	7.50	7.60	7.60	4.70	7.50	7.00	7.20	8.00
	4	8.50	7.70	7.90	7.50	4.60	7.80	7.30	7.30	8.10
	5	8.40	7.60	8.00	7.70	4.80	7.70	7.40	7.20	8.20
P2	Initial	6.30	5.80	5.50	5.80	4.80	5.50	5.80	5.80	6.30
	1	6.20	5.80	5.50	5.60	4.80	5.50	5.60	5.70	6.00
	2	6.40	5.90	5.70	5.80	4.80	5.70	5.80	5.80	6.10
	3	6.30	6.00	5.80	5.90	4.90	5.80	5.90	5.90	6.20

4	6.60	6.20	6.00	6.00	5.00	6.00	6.10	6.00	6.40
5	6.80	6.40	6.20	6.20	5.30	6.20	6.30	6.20	6.60

(Table 8)

Parameter	P1 initial	P2 initial	Difference (P1 - P2)
Calculation skill	8.0	6.3	1.7
Forecasting / pattern skill	7.5	5.5	2.0
Code / logic understanding	7.3	5.8	1.5
Noise tolerance (overloaded info)	6.8	4.8	2.0
Motivation / engagement	7.2	5.8	1.4
Fatigue / overload sensitivity	4.4	4.8	-0.4

(Table 8.5)

Agents:

Initial to Week 5 percentage growth

Agents	Arithmetic_fluency	Equation_fluency	Pattern_detection	Business_intuition	Risk_appetite	Trend_sensitivity	Conceptual_debugging	Verbal_justification	rule_following
A1	17.13%	7.80%	21.49%	15.90%	8.28%	23.47%	9.25%	12.14%	6.97%
A2	2.11%	2.18%	2.45%	18.34%	-22.04%	22.06%	6.25%	21.22%	18.42%
A3	14.02%	11.22%	3.58%	8.28%	6.26%	11.87%	5.95%	17.78%	23.00%
A4	10.13%	-0.14%	6.43%	18.54%	-10.46%	12.41%	3.20%	3.67%	36.74%
A5	21.41%	12.42%	11.68%	20.15%	-0.93%	16.49%	5.17%	5.79%	14.20%
A6	11.87%	16.21%	20.60%	3.86%	4.74%	1.00%	12.42%	-2.11%	9.58%
A7	7.93%	5.32%	23.50%	-1.39%	8.10%	4.18%	6.32%	10.32%	7.90%
A8	19.31%	9.25%	14.03%	-1.11%	2.11%	23.78%	15.39%	27.41%	10.02%
A9	6.83%	14.29%	11.40%	0.75%	10.06%	16.98%	4.69%	15.66%	-1.68%
A10	19.33%	6.77%	17.15%	10.92%	-3.16%	11.59%	14.10%	19.07%	21.41%
A11	11.75%	4.51%	13.64%	11.15%	0.99%	16.06%	-4.51%	24.32%	32.83%

A12	8.85%	4.45%	8.16%	-0.39%	16.75%	13.88%	17.91%	5.04%	25.42%
A13	11.43%	13.07%	5.92%	8.61%	8.92%	8.30%	4.64%	23.02%	20.41%
A14	16.59%	9.01%	6.52%	2.75%	-4.15%	18.73%	-4.81%	-3.77%	8.77%
A15	8.46%	11.99%	6.29%	3.74%	-2.38%	17.07%	-14.40%	16.70%	14.65%
A16	12.16%	5.08%	5.85%	11.85%	-1.25%	13.35%	0.47%	13.07%	7.60%
A17	7.25%	14.53%	8.72%	8.86%	7.14%	6.12%	27.99%	11.79%	20.07%
A18	9.60%	7.75%	8.85%	0.84%	-3.18%	12.26%	-5.84%	17.88%	10.06%
A19	9.83%	3.56%	1.50%	1.02%	-8.97%	17.93%	11.56%	21.93%	-3.51%
A20	12.68%	-0.72%	12.73%	13.10%	0.22%	14.82%	22.13%	6.00%	19.95%
A21	6.54%	10.24%	9.16%	-3.53%	10.14%	18.50%	13.19%	11.40%	3.25%
A22	7.61%	7.48%	10.71%	13.34%	4.67%	8.33%	7.66%	22.58%	24.11%
A23	14.78%	-0.75%	7.63%	18.16%	8.72%	20.43%	0.58%	9.93%	19.71%
A24	15.53%	12.20%	19.85%	14.13%	1.99%	11.42%	5.86%	36.32%	20.23%
A25	10.59%	10.05%	7.31%	11.13%	-4.50%	14.72%	33.32%	11.61%	22.21%
A26	18.58%	6.20%	10.04%	0.17%	14.16%	10.74%	-3.04%	6.48%	7.97%
A27	9.65%	5.71%	13.37%	10.53%	-8.01%	26.97%	19.12%	-7.74%	15.63%
A28	13.85%	-0.22%	9.38%	7.16%	8.68%	10.43%	10.43%	7.51%	20.30%
A29	8.81%	8.68%	7.15%	-3.68%	-9.85%	17.46%	0.68%	11.05%	7.76%
A30	5.18%	3.19%	15.26%	8.70%	2.68%	16.32%	11.41%	11.28%	14.28%
A31	15.84%	12.85%	15.03%	25.45%	-2.42%	17.84%	-2.37%	22.44%	31.79%
A32	8.28%	9.73%	1.71%	-0.56%	-4.67%	17.87%	17.49%	8.96%	7.83%
A33	10.11%	2.95%	10.88%	10.67%	-6.62%	15.05%	-4.10%	5.99%	9.79%
A34	2.72%	2.61%	13.97%	11.02%	-11.62%	7.80%	-6.39%	6.30%	23.54%
A35	15.57%	12.47%	11.23%	6.77%	5.31%	14.90%	1.36%	19.93%	4.48%
A36	11.85%	-5.26%	10.23%	20.88%	-5.62%	20.56%	8.25%	15.36%	16.81%
A37	5.65%	12.34%	13.51%	15.49%	11.16%	16.02%	-0.27%	23.94%	0.93%
A38	9.18%	5.94%	11.34%	-3.73%	4.08%	19.69%	-3.90%	12.75%	19.99%
A39	12.09%	10.04%	12.45%	6.41%	-16.52%	11.49%	2.46%	21.14%	23.38%
A40	9.68%	2.58%	10.20%	4.21%	-0.70%	14.66%	32.46%	5.50%	16.88%
A41	14.90%	3.54%	10.23%	9.57%	1.74%	23.60%	5.85%	11.83%	22.11%
A42	8.16%	3.44%	15.29%	3.88%	13.71%	4.47%	-2.30%	12.86%	19.97%

A43	9.63%	1.68%	-0.24%	-2.34%	7.41%	8.30%	-4.32%	26.72%	-6.66%
A44	8.33%	4.70%	10.18%	11.26%	-5.00%	22.10%	7.59%	13.52%	6.32%
A45	16.71%	6.50%	5.40%	2.67%	-0.99%	9.22%	-3.60%	6.60%	29.33%
A46	13.86%	6.53%	8.30%	-3.34%	-0.90%	27.06%	4.66%	3.47%	17.82%
A47	10.37%	5.95%	4.99%	8.24%	-2.52%	19.42%	10.78%	2.16%	16.50%
A48	2.84%	8.02%	4.72%	1.01%	-2.03%	7.57%	8.73%	11.68%	10.02%
A49	13.10%	8.27%	6.37%	9.41%	6.67%	9.17%	3.61%	7.35%	15.88%
A50	0.90%	1.84%	0.52%	-2.46%	-10.60%	14.47%	-3.88%	14.34%	6.46%
Average % Increase	10.99%	6.76%	9.93%	7.33%	0.51%	14.78%	6.18%	12.80%	15.03%

(Table 9)

Table of overall agent results

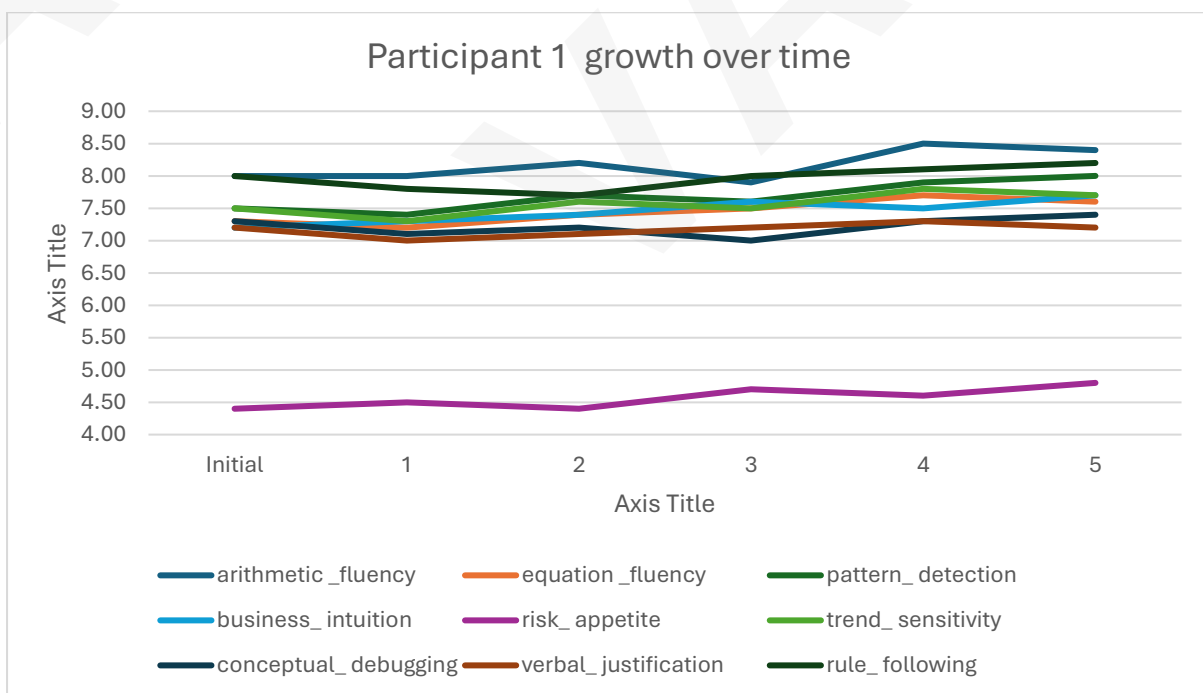
	Weeks	arithmetic_ fluency	equation_ fluency	pattern_ detection	business_ intuition	risk_ appetite	trend_ sensitivity	conceptual_ debugging	verbal_ justification	rule_ following
Mean	Initial	7.74	6.94	7.26	7.45	6.03	6.45	7.66	5.92	7.02
	1	7.76	6.94	7.32	7.44	6.03	6.45	7.66	6.01	7.09
	2	7.81	6.80	7.07	7.37	5.54	6.78	7.43	6.16	7.39
	3	8.16	7.04	7.59	7.59	5.83	7.04	7.70	6.38	7.65
	4	8.28	7.12	7.69	7.66	5.93	7.10	7.78	6.42	7.74
	5	8.60	7.41	7.99	7.99	6.16	7.38	8.11	6.67	8.01
SD	Initial	1.35	1.46	1.08	1.47	1.39	1.57	1.36	1.50	1.40
	1	1.26	1.46	1.06	1.47	1.39	1.57	1.36	1.47	1.38
	2	1.26	1.46	1.06	1.47	1.39	1.57	1.36	1.47	1.38
	3	1.52	1.53	1.24	1.53	1.64	1.65	1.38	1.72	1.39
	4	1.51	1.59	1.27	1.53	1.71	1.65	1.46	1.71	1.39
	5	1.37	1.44	1.18	1.49	1.51	1.59	1.27	1.56	1.32
Max	Initial	10.62	10.23	9.10	10.75	8.80	9.70	10.53	8.70	9.71
	1	10.00	10.23	9.12	10.75	8.80	9.70	10.53	8.73	9.72
	2	10.62	10.20	9.01	10.74	8.45	10.02	10.71	9.74	10.69
	3	11.52	10.28	9.79	11.12	9.31	10.39	10.87	10.18	10.52
	4	11.59	10.72	10.16	10.92	9.20	10.67	11.19	9.90	10.66
	5	12.27	11.15	10.89	11.69	9.58	11.28	11.50	10.62	10.94
Min	Initial	4.95	4.26	4.66	3.35	3.18	3.40	4.97	3.34	3.77
	1	5.03	4.26	4.76	3.35	3.18	3.40	4.97	3.48	3.87
	2	5.08	4.22	4.25	3.17	2.50	3.44	4.90	3.57	4.04
	3	5.20	4.31	4.59	3.07	2.50	3.75	5.19	3.54	4.27
	4	5.31	4.26	4.64	3.18	2.46	3.80	5.03	3.52	4.35
	5	5.36	4.55	4.73	3.27	2.48	3.90	5.34	3.66	4.61

(Table 10)

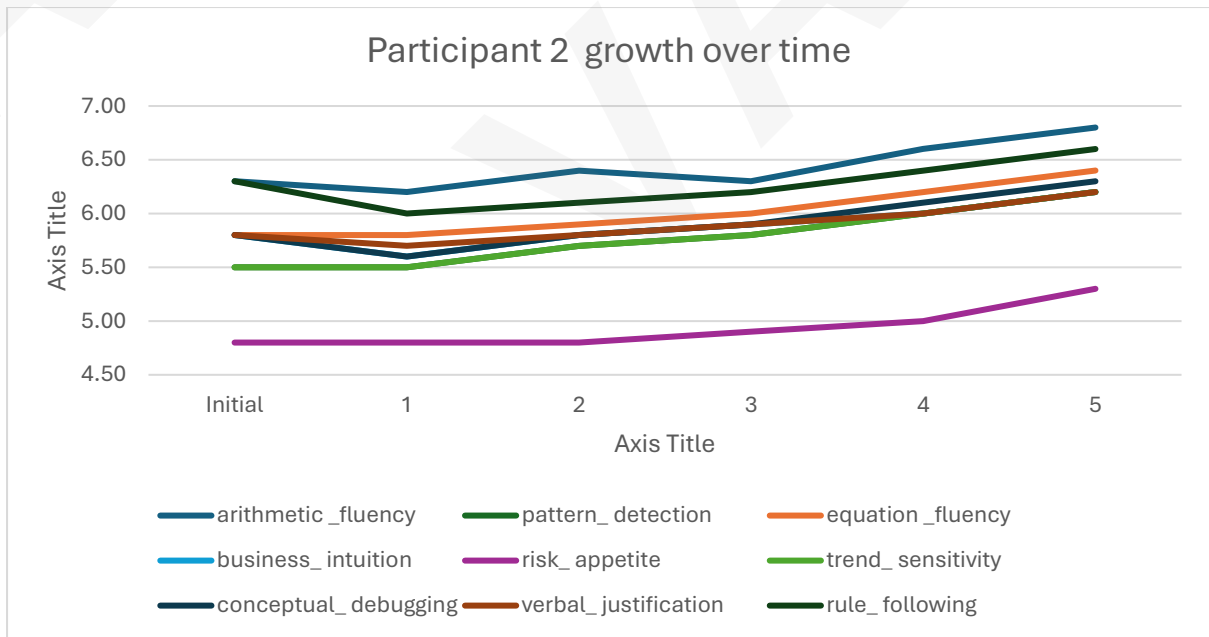
Measure	Initial Mean	Week 5 Mean	Change	Week 5 p-value
Arithmetic fluency	7.7372	8.5967	+0.8595	5.13×10^{-20}
Equation fluency	6.9370	7.4100	+0.4730	5.99×10^{-12}
Business intuition	7.4456	7.9883	+0.5427	1.97×10^{-9}
Trend sensitivity	6.4508	7.3791	+0.9283	4.12×10^{-22}

(Table 11)

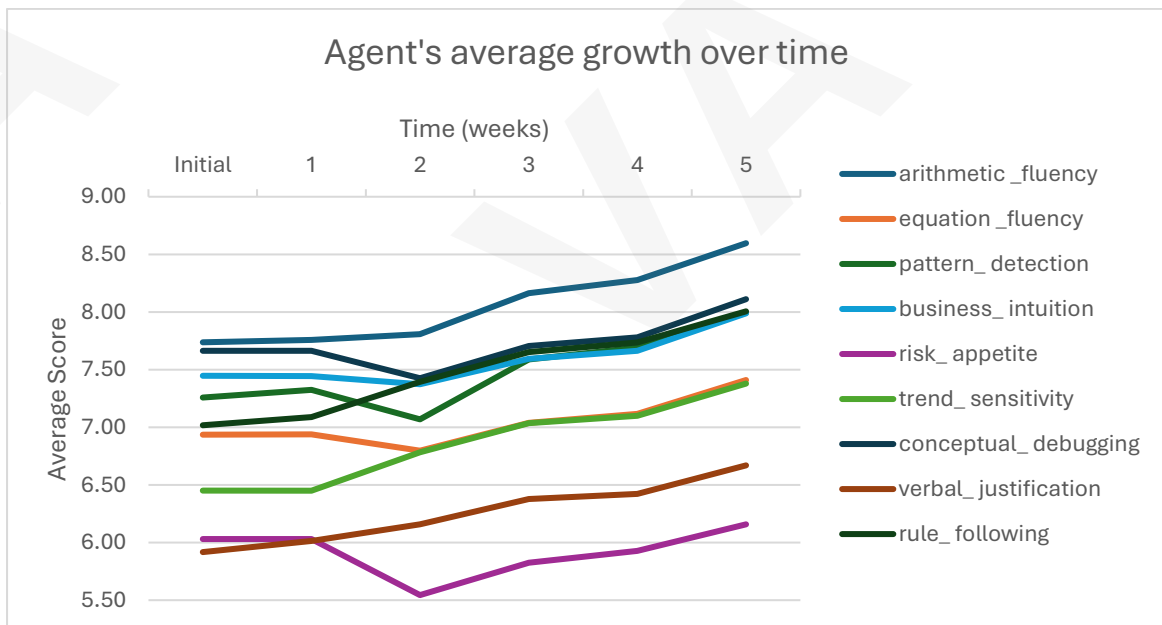
Figures:



(Figure 1.1)



(Figure 1.2)



(Figure 2)

T – Testing of results for agents:

	Initial	Week 1
t-Test: Paired Two Sample for Means		
	<i>arithmetic fluency</i>	
Mean	7.7372	7.7565208
Variance	1.861559	1.6180214
Observations	50	50

Pearson Correlation	0.995946
Hypothesized Mean Difference	0
df	49
t Stat	-0.9087
P(T<=t) one-tail	0.183979
t Critical one-tail	1.676551
P(T<=t) two-tail	0.367957
t Critical two-tail	2.009575
Effect size	-0.12851
(Figure 3.1)	

equation fluency

Mean	6.937	6.9374098
Variance	2.166903	2.1650438
Observations	50	50
Pearson Correlation	0.999998	
Hypothesized Mean Difference	0	
df	49	
t Stat	-1.03901	
P(T<=t) one-tail	0.151949	
t Critical one-tail	1.676551	
P(T<=t) two-tail	0.303898	
t Critical two-tail	2.009575	
Effect size	-0.14694	
(Figure 3.2)		

business intuition

Mean	7.4456	7.4449341
Variance	2.197058	2.1961037
Observations	50	50
Pearson Correlation	0.999998	
Hypothesized Mean Difference	0	
df	49	
t Stat	1.458419	
P(T<=t) one-tail	0.075553	
t Critical one-tail	1.676551	
P(T<=t) two-tail	0.151105	
t Critical two-tail	2.009575	
Effect size	0.206252	

(Figure 3.3)

trend sensitivity

Mean	6.4508	6.4508667
Variance	2.524897	2.5262776
Observations	50	50
Pearson Correlation	0.999999	

Hypothesized Mean Difference	0
df	49
t Stat	-0.17347
P(T<=t) one-tail	0.431499
t Critical one-tail	1.676551
P(T<=t) two-tail	0.862998
t Critical two-tail	2.009575
Effect size (Figure 3.4)	-0.02453

T – Testing of results for agents:

	Initial	Week 5
t-Test: Paired Two Sample for Means		
	<i>arithmetic fluency</i>	
Mean	7.7372	8.59670346
Variance	1.86155935	2.601899394
Observations	50	50
Pearson Correlation	0.97704803	
Hypothesized Mean Difference	0	
df	49	
t Stat	-15.0603268	
P(T<=t) one-tail	2.5643E-20	
t Critical one-tail	1.67655089	
P(T<=t) two-tail	5.1287E-20	
t Critical two-tail	2.00957524	
Effect size (Figure 4.1)	-2.12985184	

	<i>equation fluency</i>	
Mean	6.937	7.41003609
Variance	2.16690306	2.710640729
Observations	50	50
Pearson Correlation	0.97774735	
Hypothesized Mean Difference	0	
df	49	
t Stat	-8.99549293	
P(T<=t) one-tail	2.9938E-12	
t Critical one-tail	1.67655089	
P(T<=t) two-tail	5.9876E-12	
t Critical two-tail	2.00957524	
Effect size	-1.27215481	

(Figure 4.2)

<i>business intuition</i>		
Mean	7.4456	7.988329561
Variance	2.1970578	2.715212551
Observations	50	50
Pearson Correlation	0.94965879	
Hypothesized Mean Difference	0	
df	49	
t Stat	-7.3406951	
P(T<=t) one-tail	9.8559E-10	
t Critical one-tail	1.67655089	
P(T<=t) two-tail	1.9712E-09	
t Critical two-tail	2.00957524	
Effect size	-1.03813106	

(Figure 4.3)

<i>trend sensitivity</i>		
Mean	6.4508	7.379059709
Variance	2.52489731	3.063584905
Observations	50	50
Pearson Correlation	0.97765193	
Hypothesized Mean Difference	0	
df	49	
t Stat	-16.9288175	
P(T<=t) one-tail	2.061E-22	
t Critical one-tail	1.67655089	
P(T<=t) two-tail	4.1219E-22	
t Critical two-tail	2.00957524	
Effect size	-2.39409632	

(Figure 4.4)

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